Manoj G

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Professional Summary

Computer Science student building secure software, AI systems, and developer tools. Experienced in backend engineering and AI agents, focused on scalable, production-ready software.

Education

PSG Institute of Technology and Applied Research, Coimbatore 2023–2027
– B.Tech Computer Science & Business Systems **CGPA: 8.20 / 10**

Government Higher Secondary School, Namakkal 2023
– Higher Secondary Education (XII) **Percentage: 92.33%**

Experience

Froze Communication Jan 2026 – Jun 2026
Project Intern Remote

- Built and deployed a **multi-tenant endpoint detection and response (EDR) platform** that collects Windows Event Logs, detects brute-force attacks, and centralizes threat monitoring across multiple organizations.
- Engineered a **Windows endpoint agent** for Windows Event Log collection, event deduplication, firewall synchronization, and auto-updates, supporting reliable batched event ingestion.
- Designed a **threat detection and response pipeline** with automated IP blocking, ˜30s firewall synchronization, threat intelligence enrichment, and real-time SSE alerts for rapid threat visibility and mitigation.

Projects

0xMemory – AI Agent Memory Layer *Python, FastAPI, ChromaDB, MCP* |

- Built a **local-first memory layer for AI agents** using the Model Context Protocol (MCP), enabling persistent, project-scoped context across sessions without cloud dependency.
- Designed a **dual-storage architecture** syncing Markdown files with a ChromaDB vector index, enabling hybrid keyword and semantic search with fully local data storage.
- Implemented an **LLM-powered knowledge extraction pipeline** to convert raw chat logs into structured facts, enabling efficient memory recall and reduced token usage.

0xNexigent – AI Runtime FinOps Gateway *Redis, AWS S3, PostgreSQL, React* |

- Built an **intercepting AI proxy gateway** enforcing multi-tier hierarchical budget governance across Teams, Agents, and Sessions to prevent financial exposure from runaway LLM calls.
- Engineered an **atomic budget reservation engine** using Redis Lua scripts for single-pass holds, microdollar financial reconciliation, and sliding-window runaway circuit breakers.
- Implemented an **AWS S3 audit storage layer** for offloading large prompt/response payloads and enabling timeline-based audit replay, reducing PostgreSQL storage and query overhead.

Technical Skills

Languages: Python, C#, JavaScript, SQL
Backend: FastAPI, ASP.NET Core, Noje.js, REST APIs
Databases & Cloud: PostgreSQL, ChromaDB, AWS (S3, EC2)
Tools: Linux, Git, GitHub Actions
Security: Secure Authentication (JWT, RBAC), Threat Detection, API Security
AI/LLM: Model Context Protocol (MCP), AI Agents, RAG, Prompt & Context Engineering

Open Source & Certifications

- **Open Source:** Contributed to **OWASP cve-lite**, improving static dependency analysis accuracy through scanner enhancements and automated regression testing.
- **Certification:** Google Cybersecurity Professional Certificate (Coursera).